

Large-Scale Education Reform in General Equilibrium: Regression Discontinuity Evidence from India

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The economic consequences of large-scale government investments in education depend on general equilibrium effects in both the labor market and the education sector. I develop a general equilibrium model capturing the consequences of massive countrywide schooling initiatives. I provide unbiased estimates of the model's elasticities, using a regression discontinuity derived from Indian government policy. The earnings returns to a year of education are 13.4%, and the general equilibrium labor market effects substantially depress returns, by 6.6 percentage points. These general equilibrium effects have distributional consequences across cohorts and skill groups, whereby unskilled workers are better off and skilled workers worse off.

Large-scale education expansions represent substantial investments of public resources and benefit households by increasing local productivity. However, since they affect both individual behavior and labor market outcomes, convincing causal estimates of their overall economic benefits are

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hard to generate. While small-scale, carefully controlled, researcher-led experiments provide promising evidence about which educational investments are effective, these estimates may not be sufficient when evaluating large-scale policies. Importantly, large-scale education programs may have sizable general equilibrium (GE) effects on both education and labor markets. I estimate and take into account these GE effects while determining the overall economic implications of nationwide education programs.

I build a framework to analyze the consequences of a large-scale educational expansion program in India with an explicit focus on issues inherent to nationwide government policies: the GE effects in the markets for both education and labor. I model the labor market and the education sector and decompose wage changes into the individual returns to education and GE effects, which together determine welfare consequences. On the labor market side, I combine models of education choice (Becker 1967) with determinants of the skill premium (Card and Lemieux 2001) to study how the distribution of earnings affects education choices, and consequently how changes in education affect the distribution of earnings.

The allocation rule under which Indian districts receive the funding allows me to estimate important elasticities using a regression discontinuity (RD) approach. The policy raised education levels for the young. Consequently, I further exploit variation between younger (young enough to change education) and older cohorts and between high-skill (educated) and low-skill workers to identify GE effects, by estimating how the earnings skill premium changes across local economies. I measure the overall benefits of the policy and its distributional consequences across skill levels and age cohorts. Not only do I find substantial GE effects in the labor market but I am also able to precisely estimate their size—these effects depress the returns to skill by 33% and dampen the increase in labor market benefits by 23.8%. By expanding the skilled workforce, the policy makes skilled workers worse off and unskilled workers better off and leads to the adoption of skill-biased capital. At the same time, the GE effects in the education sector suggest a crowd-in of private schools, negating concerns of crowd-out.

As education levels rise, we expect earnings and therefore the returns to be affected in a few ways. First, more educated workers are more productive and earn higher wages. Second, educated workers may reside in regions where there are fewer educated workers, making them relatively more

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valuable in the labor market. But if large numbers of people receive additional education, there is also a GE effect in the labor market: an increase in the abundance of high-skill labor puts downward pressure on the earnings skill premium. Yet as more skilled workers join the labor force, skill-biased capital may be adopted by firms in these regions, raising the premium. Indeed, as workers switch to more productive skill groups, overall output may increase, to the benefit of all workers. I estimate all of these components of the GE effects to better quantify the distributional impacts and the changes in labor market benefits.

The policy I study was India's flagship education scheme in the 1990s and early 2000s, the District Primary Education Program (DPEP), which expanded public schooling in half the country by targeting low-literacy regions. At that time, it was the largest program for primary education in the world, in terms of geography, population, and funding, suggesting that its consequences would be similarly widespread (Jalan and Glinskaya 2013). Such schooling expansions reduce the marginal cost of attaining education by improving access to schools (Birdsall 1985; Behrman, Birdsall, and Kaplan 1996; Duflo 2001), inducing some students who have potentially high returns to schooling, but could not previously afford it, to get more education.

Under the allocation rule, districts that had a female literacy rate below the national average were more likely to receive the program. I compare regions on either side of the cutoff to estimate causal impacts. The RD allows me to tackle biases that arise when estimating the individual returns to education and when comparing earnings in two different local economies. I compare students induced into getting more education to similarly competent students who were not. At the regional level, the RD tackles biases that arise as a result of differences in the local economy and labor market, as some regions may have more skilled workers or industries.

To support each piece of the GE model, I create a comprehensive data set, combining three waves of a household survey, a census of firms, school-level data, test score surveys, and the Indian census. I assemble a 10-year panel of districts to track long-run outcomes, allowing me to follow local labor and education markets over time.

I use the data to estimate the returns to education and the GE effects, exploiting not just the RD but also the variation in cohort exposure and skill levels. Younger cohorts can change their educational attainment in response to the policy, whereas older cohorts cannot. However, the young and the old are differently affected by changes in the labor market skill distribution. I estimate the earnings skill premium by age group separately on either side of the RD cutoff. The difference in the earnings skill premium for older workers allows me to measure the GE effects for all cohorts. At the same time, since the young and old are not perfect substitutes (and in some contexts may be complements), there is an often-overlooked additional impact on young workers, which I estimate by looking at the additional change

in the skill premium for the young. Using the estimated parameters, I measure the overall impact of the policy on welfare for the different types of workers and cohorts.

Given evidence from other contexts, it is important for us to address such labor market effects. In the United States, Heckman, Lochner, and Taber (1998a, 1998b, 1998c), Lee (2005), and Abbott et al. (2019) show how changes in taxes or tuition and financial aid may have large GE effects.¹ I flexibly model and estimate the GE effects on different cohorts and skill groups. This allows me to determine distributional consequences across both dimensions and estimate crucial economic parameters and schooling returns in both the presence and the absence of GE effects.

I corroborate my results with a difference-in-differences (DID) analysis, where I compare treated to untreated districts and younger to older cohorts. However, as I show, using a DID design, it is challenging to recover the GE effects and, accordingly, adjust the returns to skill. Duflo (2004) shows that Indonesia's school building depressed average wages for older, untreated cohorts. As the young and old differ in skill composition and are not perfect substitutes (and, in fact, may be complements), I show that the old may not be a reasonable counterfactual for the young when there are GE effects.

The advantage of the RD is that I can estimate the entire extent of GE effects and disentangle them into the portion that affects all cohorts and additional impacts on treated cohorts. This is because, in the RD, I do not need to rely on the effects on the old to identify the effects on the young. Yet simply comparing the change in average earnings of the young across the RD cutoff (for a unit increase in education) does not recover either the partial or GE returns to education, as one may do with a conventional IV (instrumental variable)-Wald estimator. I estimate the GE effects on each skill group separately and for all cohorts (including treated cohorts), thus allowing me to separately identify the returns to education in both partial and general equilibria.² Indeed, I find that the GE effects on the old are negligible in comparison to the GE effects on the (treated) young. The impact on older cohorts is an object of interest on its own, and such spillovers may be missed if we focus on treated cohorts alone.

I find that the program increased both education and earnings for students in targeted regions. I find large overall economic benefits to households, driven by reductions in the costs of education and increases in the overall output of the region. However, GE effects substantially mitigate the rise in labor market earnings for those who acquire more skill. Increases in

¹ Labor market GE effects include job assistance programs (Crépon et al. 2013), calibrated macro models (Albrecht, van den Berg, and Vroman 2009), effects on demographics (Epple and Ferreyra 2008), and major choice (Bianchi 2020).

² Given elasticities of complementarities across skill and cohorts, the GE effects may also raise average wages for older cohorts as unskilled wages rise.

the supply of educated workers dampened earnings for skilled workers and put upward pressure on the earnings of unskilled workers. The returns to a year of education, including GE effects, are 13.4%. But the estimated labor market GE effects are substantial—for a 17 percentage point increase in the fraction of skilled workers, the GE effects depress the returns by 6.6 percentage points and dampen the increase in benefits to students by 23.8%.³ These GE effects have distributional consequences, with a transfer of labor market benefits from skilled to unskilled workers, particularly among the young. High-skill workers who did not change their education under the policy are adversely affected, whereas low-skill workers benefit.

My analysis allows for both the mobility of workers across regions and the adoption of skill-biased capital or technology. Importantly, the adoption of skill-biased capital does play a role, however small, in mitigating the GE effects. But consistent with the other literature in this context (Munshi and Rosenzweig 2009, 2016), I find no evidence of labor mobility.

These results have significant implications for research and policy. First, there are few contemporary estimates of skill returns from developing countries such as India, which was undergoing a sustained period of growth in both education and income. Second, my methods broadly speak to our attempts to estimate the (causal) private returns to education using macro-level variation from tuition reductions, compulsory schooling laws, schooling expansions, or large-scale policy reforms. Such large-scale variation identifies a different parameter, as it is no longer one person being treated with a year of education, as envisioned by Mincer (1958), Becker (1967), and Willis (1986). Rather it involves treating an entire cohort of students, leading to GE effects, which I find to be substantial. Third, I show that returns to education are not fixed parameters but rather endogenous quantities that depend on the local labor market, and I derive meaningful relationships between labor market changes and returns.

Fourth, a substantial number of microinterventions help guide policy makers.⁴ My analysis complements these with information on how to measure the effects of scaled-up versions of such interventions.⁵ The GE effects

³ As the policy targets the upper primary level and the largest education response is there, these returns are largely representative of a year of upper primary schooling. Given the focus on primary schools (under the Millennium Development Goals and national efforts such as DPEP), this is a policy-relevant parameter.

⁴ In India, these studies cover library programs (Borkum, He, and Linden 2012), teacher incentives (Muralidharan and Sundararaman 2010), bicycles (Muralidharan and Prakash 2017), computer-aided programs and remedial education (Banerjee et al. 2007), and teacher quality or absence (Duflo, Hanna, and Ryan 2012; Muralidharan et al. 2017).

⁵ While the evidence on smaller changes of inputs within schools is mixed (Muralidharan 2013), large-scale investments in schooling like the one studied here have been found to be relatively more successful across the world. Some examples are in Indonesia (Duflo 2001), Burkina Faso (Kazianga et al. 2013), Zimbabwe (Agüero and Bharadwaj 2014), Nigeria (Osili and Long 2008), Uganda (Deininger 2003), Zambia (Ashraf et al. 2020), Kenya (Bold, Kimenyi, and Sandefur 2013), and India (Chin 2005; Afridi 2010; Adukia 2017).

will either undermine or amplify the effectiveness of microinterventions when they are scaled up (Acemoglu 2010; Deaton 2010; Muralidharan and Niehaus 2017; Egger et al. 2019). This point is stressed even outside the realm of development economics (Heckman, Lochner, and Taber 1998a, 1998b; Heckman, LaLonde, and Smith 1999).

Finally, these methodological insights can be applied in other contexts, allowing researchers and policy makers to estimate the welfare consequences of such interventions around the world (e.g., school building, compulsory schooling, fee reductions, or job programs).⁶

I. DPEP

I derive plausibly exogenous variation from a large schooling expansion policy in India, where any district below the 1991 national average female literacy rate was eligible to receive funds. There are two advantages of using districts around the national average. The first is that there is a large density of districts at the RD cutoff, and the second is that this analysis is representative of the district with average literacy induced into receiving the policy.

In 1994, DPEP was introduced, eventually serving 271 of approximately 600 districts in the country. DPEP grew with funding from international agencies, making it one of the largest donor-assisted programs in the world (Jalan and Glinskaya 2013).⁷ In 2002–3 alone, \$345 million in foreign funds was spent, concentrated in less than half the districts in the country, allowing for a valuable policy experiment. It was the flagship education program for more than a decade but was phased out in 2006.⁸

The program served approximately 51.3 million children. The affected districts were geographically dispersed across the country (map in fig. A.1). It created about 160,000 new schools, trained 1.1 million teachers and 3 million community members, and increased funds for primary school education by between 17% and 20%. The primary objective was to improve student access to and retention in primary and upper primary education. Numerous policy briefs and media reports highlight the program's success.⁹ Other work shows that DPEP increased education in treated districts (Jalan and Glinskaya 2013; Azam and Saing 2017). Over the period, districts could received about \$8 million, or \$9.1 per student, and even in the short run this intervention lowered private household costs of schooling by

⁶ The RD is simply a special case of an IV method. The insights here can be applied to any IV that leverages variation from a large-scale policy.

⁷ States maintained the level (in 1992; Varghese 1994) and growth rate (World Bank 1997) of expenditure that existed before the program, to ensure no crowd-out of funds. Taxes were not raised to directly fund DPEP.

⁸ Source: Parliamentary Questions 1807, 552, 55, 267, 1320, and 2018 and Upper House Question 2855.

⁹ See World Bank (2003), *Economic Times* (2005), and Government of India (2011).

20%–40% (Jalan and Glinskaya 2013). Additional details on the history, funding, and secondary objectives can be found in appendix C.

II. Model

My framework captures salient features of the local economy and market for education, including GE effects of policy reforms. I derive estimation equations and identify elasticities that determine the effect of schooling expansions on economic benefits in section IV.¹⁰

A. Economic Production and the Labor Market

Aggregate output Y_d in district d depends on L_d (effective labor) and K_d (capital).¹¹ Capital is perfectly elastically supplied across districts at rental rate R^* .¹² Effective labor supply L_d depends on the labor aggregate L_{sd} at each skill level s .

$$Y_d = L_d^\varrho K_d^{(1-\varrho)}, \quad \text{where } L_d = \left(\sum_s \theta_{sd} L_{sd}^{(\sigma_E-1)/\sigma_E} \right)^{\sigma_E/(\sigma_E-1)}, \quad 0 < \varrho < 1, \quad (1)$$

is the share of output accruing to labor, $\theta_{sd} > 0$ captures worker productivity with education or skill level s , and $\sigma_E > 0$ is the elasticity of substitution across education or skill groups. The productivity parameter θ_{sd} captures the productivity of each skill level and increases with an increase in skill-biased capital in the district k_{sd} , such that $\theta'_{sd}(k_{sd}) > 0$.¹³ The value of θ_{sd} therefore varies across districts only because of the variation in skill-biased capital k_{sd} . The aggregate supply of workers at skill level s depends on the aggregate effective supply of workers in each skill level ℓ_{asd} in a given age cohort a :

¹⁰ As I explain in sec. IV.B, the advantage of relying on only a few elasticities is that the estimation procedure and measured GE effects do not directly depend on imposed functional forms. Indeed, the estimated wage benefits will hold true even under many alternative formulations. However, couching it in a canonical labor economics model allows us to understand the parameters driving the effects, drawing links to important elasticities estimated in other work and underlying determinants of the returns to education.

¹¹ Including nontradables, such as land, in the aggregate production function does not directly affect estimation. The policy will theoretically change the value of nontradables, yet I focus on the earnings of workers, and not returns to owners of capital and land.

¹² The assumption of perfectly elastic capital is not essential. The results are unaffected by assuming a fixed capital stock (see app. B.2).

¹³ The amount of “unbiased” capital does not affect the returns to skill, yet skill-biased capital does. When there is an increase in the supply of skilled workers, we may expect the skilled wage to fall. Yet if firms start using more skill-biased capital (now that they have skilled labor), then that mitigates the fall in skilled wages. Allowing for such demand-side responses will allow for dampened GE effects. For completeness, in app. B.3, I explicitly model skill-biased capital within the nested CES (constant elasticity of substitution) framework and show how flexible ways of incorporating it do not affect the estimation or results.

$$L_{sd} = \left(\sum_a \psi_a \ell_{asd}^{(\sigma_A-1)/\sigma_A} \right)^{\sigma_A/(\sigma_A-1)}. \quad (2)$$

Here, σ_A is the elasticity of substitution across age cohorts and ψ_a is the productivity of a specific cohort. The effective supply ℓ_{asd} can depend on the ability of workers ϵ_{is} and a worker gets paid their marginal product.¹⁴ The average log earnings are therefore

$$\begin{aligned} \log w_{asd} &= \log \left(\frac{\partial Y_d}{\partial \ell_{asd}} \right) = \log \tilde{\varrho} + \log \theta_{sd} + \log \psi_a \\ &\quad + \frac{1}{\sigma_E} \log Y_d + \left(\frac{1}{\sigma_A} - \frac{1}{\sigma_E} \right) \log L_{sd} - \frac{1}{\sigma_A} \log \ell_{asd}, \end{aligned} \quad (3)$$

where $\log \tilde{\varrho} \equiv [1 - (1/\sigma_E)][(1 - \varrho)/\varrho] \log((1 - \varrho)/R^*)$ is common across all districts and workers.¹⁵

There are a few components that drive differences in average earnings when comparing two different types of people in two different labor markets, as in equation (4):

$$\begin{aligned} \log \left(\frac{w_{asd}}{w_{d's'd'}} \right) &= \underbrace{\log \left(\frac{\theta_{sd}}{\theta_{d's'd'}} \right)}_{\text{productivity}} + \underbrace{\log \left(\frac{\psi_a}{\psi_{a'}} \right)}_{\text{cohort}} \\ &\quad + \underbrace{\frac{1}{\sigma_E} \log \left(\frac{Y_d}{Y_{d'}} \right)}_{\text{output}} + \underbrace{\left(\frac{1}{\sigma_A} - \frac{1}{\sigma_E} \right) \log \frac{L_{sd}}{L_{d's'd'}}}_{\text{skill distribution}} - \underbrace{\frac{1}{\sigma_A} \log \frac{\ell_{asd}}{\ell_{d's'd'}}}_{\text{skill-cohort distribution}}. \end{aligned} \quad (4)$$

This equation is crucial, in that it captures why earnings are systematically different across people and labor markets. In the first component, productivity, θ_{sd} is the higher productivity associated with more education. Not only are skilled workers more productive, but variation in the supply of skill-biased capital across districts also affects earnings. The second component, cohort, captures age-specific productivities and returns to experience ψ_a . The third, output, is the difference across labor markets related to differences in the size of the economy. The fourth, skill distribution, is the difference in earnings due to differences in the supply of more educated workers, L_{sd} . This influences the labor market GE effects that affect all cohorts. The last, skill-cohort distribution, affects earnings due to differences in the supply of skilled workers within each cohort ℓ_{asd} and drives an additional GE

¹⁴ In sec. V.C.6, we examine the implications of labor market distortions, factor misallocation, and monopsonistic power.

¹⁵ This is at the optimal value of K_d^* , where $Y_d = [(1 - \varrho)/R^*]^{(1-\varrho)/\varrho} L_d$. For tractability, I ignore the role of changing prices of nontradables. One can include a $\log P_d$ with the $(1/\sigma_E) \log Y_d$ term, not affecting the returns to skill.

effect on cohort a . Changes in the skill distribution by age will have important GE effects on earnings.

How much the skill distribution affects the difference in earnings depends on the elasticities of substitution σ_E and σ_A . For instance, if the young and the old are perfect substitutes, then the skill-cohort distribution should not affect earnings. The increase in earnings for a person who goes from being unskilled u to skilled s will be defined as the returns to skill β_{asd} :

$$\log \frac{w_{asd}}{w_{aud}} = \log \frac{\theta_{sd}}{\theta_{ud}} + \left(\frac{1}{\sigma_A} - \frac{1}{\sigma_E} \right) \log \frac{L_{sd}}{L_{ud}} - \frac{1}{\sigma_A} \log \frac{\ell_{asd}}{\ell_{aud}} \equiv \beta_{asd}. \quad (5)$$

This highlights an important fact: the returns to skill are not an exogenous parameter but rather an endogenous variable that depends on local labor market conditions: the difference in the productivity parameters θ_{sd} and θ_{ud} , the skill distributions L_{sd} and L_{ud} , and the cohort-specific skill distributions ℓ_{asd} and ℓ_{aud} . In regions that have relatively more skilled workers, the returns to acquiring skill will be relatively lower, whereas for regions with more skill-biased capital, the returns are higher.

Importantly, as I show in appendix B.6, migration is consistent with this setup. If migration is not skill biased, it will not affect the relative quantities in equation (5). However, skill-biased migration will change these quantities and affect skill premia. Theory suggests that such flows may be in the direction of equalizing wages, attenuating the negative GE effects.

B. Students' Decisions

Students choose the optimal level of education, given their marginal costs of schooling and returns to education (Mincer 1958; Becker 1967; Willis 1986). Given how earnings are determined in section II.A, these choices will also eventually affect earnings and lifetime utility.¹⁶ Student i chooses their years of education s_{id} to maximize the present discounted value of expected log lifetime earnings $\mathbb{E}[\log w_{aid}(s_{id})]$, given schooling costs $\kappa(s_{id})$:¹⁷

$$\max_{s_{id}} \mathbb{E}[\log w_{aid}(s_{id})] - \left(\log r_{id} + r_{id}s_{id} + \frac{1}{2} \Gamma s_{id}^2 \right), \quad (6)$$

¹⁶ Willis (1986) shows how such formulations are derived from conventional utility functions. Intuitively, maximizing lifetime earnings maximizes lifetime consumption, and thereby utility when consumption is optimally allocated across periods.

¹⁷ Since the linear form of $\kappa(s_{id})$ captures only the opportunity costs, Card (1999) suggests a more general formulation of the cost function, to capture credit and other monetary constraints. Becker (1967) suggests the quadratic costs from the observation that each subsequent year of education is even more expensive than before, as (a) fees are higher for higher levels (and in many cases early education is subsidized) and (b) students first exhaust easy sources of funds (parents, relatives) before using more expensive sources (loans).

where Γ is the quadratic cost parameter. Equations (4) and (5) determine the form of the individual earnings function. The return to an additional year of schooling $\tilde{\beta}_{asd}$ depends on returns to skills β_{asdb} and the difference in schooling years between the skilled (s_1 years) and the unskilled (s_0), where $\tilde{\beta}_{asd} = \beta_{asd}/(s_1 - s_0)$. These returns differ across districts, cohorts, and skill:¹⁸

$$\mathbb{E}[\log w_{aid}(s_{id})] = \mathbb{E}[\gamma_d + \gamma_a] + \mathbb{E}[\tilde{\beta}_{asd}]s_{id} + \log \epsilon_i. \quad (7)$$

Unlike the aggregate average wage function in equation (3), the individual-level earnings also depend on ϵ_i , the ability of the worker (not observable to researchers), the distribution of which is the same across districts. This ability is correlated with the marginal costs of schooling r_{id} and leads to biases in standard OLS (ordinary least squares) regressions ($\text{corr}(\epsilon_i, r_{id}) \neq 0$); high-ability workers earn high wages but have lower costs of schooling. Crucial to note is that the returns to education $\tilde{\beta}_{asd}$ differ across districts and skill groups because of differences in relative skills in the local labor force and across cohorts because of the cohort-specific differences in the skill distribution. Unlike equation (3) on aggregate returns, here there is an expectations operator, as schooling decisions respond to expected, rather than actual, returns (Jensen 2010; Hastings, Neilson, and Zimmerman 2018).

In equation (7), average earnings also differ across districts γ_d because of differences in the overall output and capital across regions, and they differ across age cohorts γ_a , given returns to experience or other cohort-specific productivities captured in equation (4). From the first-order conditions, one can obtain the optimal years of education for person i :

$$s_{id}^* = \frac{\mathbb{E}[\tilde{\beta}_{asd}] - r_{id}}{\Gamma}. \quad (8)$$

The variation in s_{id}^* within a district d is driven entirely by the variation in the marginal cost parameter r_{id} .¹⁹ The marginal cost parameter for person i in district d is a function of the district-level costs of going to school and an individual component η_i that captures individual heterogeneity in the costs of schooling. The district-level costs depend on the

¹⁸ For tractability, I suppress any time subscripts. The expectations operator captures expectations on future earnings at the time the schooling decisions are made. Note that at the time of schooling decision, schooling levels and abilities are known and hence have no expectations operator. Yet there may be uncertainty on future earnings by skill, cohort, and district.

¹⁹ However, the distribution of earnings is driven both by the costs of education r_{id} and by abilities ϵ_i . Smith (1775, 29) highlights the importance of educational capabilities r_{id} when arguing that “the difference between the most dissimilar characters, between a philosopher and a common street porter, for example, seems to arise not so much from nature, as from habit, custom and education.” On the other hand, early models of variation in earnings (Roy 1951) discuss the importance of “abilities” ϵ_i , such as “health, strength, skill, and so on” (135).

access to schooling A_d (distance to the nearest school) and the monetary price of school p_d (such as school fees),

$$r_{id} \equiv -\Psi A_d + p_d + \eta_i, \quad (9)$$

where Ψ represents how access to education affects individual i . Having more schools in regions that did not initially have many lowers transportation costs of going to school but may also lower competitive equilibrium fees, even for private schools. Consequently, the individual costs of schooling are a direct function of the funds received under the program, R_i .

The schooling response in equation (8) also depends on expected wage returns $E[\beta_{asd}]$. It is not necessary that households have perfect foresight over their expected wage, for instance, as they may be unable to predict the magnitudes of the GE effects. If expectations (whether accurately or not) suggest substantially low returns, then cost reductions may not induce many to get education. Consequently, expectations will affect how a fall in the costs of schooling increases education (Heckman, Lochner, and Taber 1998a, 1998b, 1998c).²⁰ Yet some of the later analysis is focused on examining the change in ex post education returns as skill prices adjust in response to a given change in schooling. Here, expectations matter less. As I show below, when measuring the economic benefits, once again, expectations play an important role.

In appendix B.6, I expand the model to include migration decisions. This is a period in India with well-documented evidence of low migration rates despite wide regional wage disparities (Munshi and Rosenzweig 2009, 2016; Topalova 2010), and empirically I fail to detect a migration response.²¹ Theoretically, migration may affect how schooling subsidies translate into an increase in the size of the skilled workforce. Yet for much of the analysis, I examine how a given increase in the size of the skilled workforce reduces the skill premium. If skilled wages fall, then skilled workers may emigrate, muting the outward shift in the supply of skilled workers and mitigating any GE effects. Consequently, the GE effects are for a given change in the skill distribution. In the model extension detailed in appendix B.6, migration is an individual decision and varies across cohorts and education level. The extension clarifies that migration changes the slope of the labor supply curve, even as one of my primary aims is to estimate the slope of the (relative) labor demand curve for a given shift in labor supply.

²⁰ If expectations adapt slowly, we may even generate interesting dynamics in future cohort schooling decisions, such as cobweb-style models (Freeman 1975). Indeed, students in India seem to react to expected returns (Khanna and Morales 2017; Adukia, Asher, and Novosad 2020; Khanna 2020).

²¹ Many studies on India are explicit about ignoring migration in the main analysis, as the numbers are low (Deshingkar and Anderson 2004; Anderson 2005; Duflo and Pande 2007; Banerjee, Iyer, and Somanathan 2008).

In appendix B, I have a detailed discussion of the supply of both public and private schooling (app. B.1), and the model's equilibrium is described in appendix B.4. Public schools depend on funds received from the government, R_d , and expand the supply of schooling in response. Private schools make an entry or exit decision on the basis of expected profits. If the expansion of public schooling lowers equilibrium school fees, p_d , private schools may be crowded out. Alternatively, if demand externalities raise the demand for private schooling, or if public expansions lower entry costs for private schools, they may be crowded in (app. B.1).

III. Data

I combine a number of large data sets, merged at the district level, which is the relevant local economy and labor market in this context (Duflo and Pande 2007). I combine data on school-level inputs; household-level data on education, migration decisions, and schooling expenditures; labor market data on earnings and occupations; and firm-level data on types of manufacturing in the different regions. Data details can be found in appendix D.

Data for school construction and inputs come from the District Information System for Education (DISE), covering about 1.45 million schools by 2012. I compile data at the school level for all waves between 2005 and 2012, including the number of schools, when they were built, whether public or privately owned, the number of teachers by education level, and infrastructure. Table A.1 summarizes variables in 2005, at the end of DPEP. Twenty-seven percent of the schools in 2005 were built after 1993, 20% government owned and the remaining 7% private schools.

To study educational outcomes, I use household surveys and census data. I create a panel of districts, using the 1991 census as a baseline. District splits and merges are well documented by official census crosswalks, so I normalize changes to districts to be at the 1991 level. The 1991 census female literacy rate, used by policy makers, is the running variable for the RD.

I use three rounds of the National Sample Survey (NSS), between 2004 and 2010, to examine impacts on education, earnings, expenditures, migration, and other labor market characteristics. It is the largest nationally representative household survey in the country, asks questions on weekly activities for up to five different occupations per person, and records weekly earnings for each individual in the household. It also includes a detailed expenditures module that includes educational expenditures. Summary statistics for the 2009 NSS round are presented in table A.2. In 2009, only about 66% of the population had finished primary school, and on average, people had 6.8 years of education and earned about \$37

a month. The 2009 NSS round is the first large-sample labor force round after the end of DPEP, and it has the added advantage of allowing time for students affected by the policy to enter the labor market. I restrict individuals to be between 17 and 75 years of age, but results are robust to relaxing this constraint.

To study firm behavior, I use a census of manufacturing firms from the Annual Survey of Industries (ASI) and obtain the district (gross) domestic product from each state's statistical office. Finally, I use the Annual Status of Education Report (ASER) to study impacts on test scores. The ASER is collected annually by a nongovernmental organization (Pratham) and surveys children between the ages of 3 and 16. These surveys are done at home on weekends, so as to capture school dropouts and those who never attended school. I use all available surveys between 2007 and 2012.

IV. Estimation and Identification Using an RD

The DPEP targeted low-literacy districts. Districts that had a female literacy rate below the national average (based on the 1991 census) were eligible for the program. However, not all such districts were selected, generating a fuzzy RD design using the 1991 female literacy rate as a running variable. To my knowledge, no other programs use the district-level 1991 female literacy rate as a cutoff.

As the RD cutoff was around the national average, there is a large density at the cutoff (fig. 1A), allowing for robust identification. Consequently, my estimates are representative of a (policy-relevant) average-literacy district induced into taking up treatment. Since we should not expect any discontinuity in the baseline distribution of individual labor market abilities or costs of schooling around the cutoff, the RD estimator is consistent. Indeed, at the cutoff, we expect no discontinuity in prepolicy labor market characteristics, skill-biased capital, or regional output. I show that cohorts that were too old to change their schooling by the time the policy was implemented have no discontinuity in educational attainment. In order to estimate the GE effects, I further exploit variation in cohort exposure and skill levels.

Since more able workers may also be more capable students, OLS estimates are biased, and the variation generated by the policy overcomes this bias. The policy induces certain students to go to school, while identical students in nonpolicy regions do not. Following students into the labor market, I compare wages in the two regions to determine the returns to schooling for the subpopulation that was induced into getting more education. At the same time, local labor markets may differ widely across regions in terms of their skill distributions and skill premiums. This will confound OLS estimates of the GE effects. The RD allows me to compare similar local economies that differ only on the access to the DPEP policy.

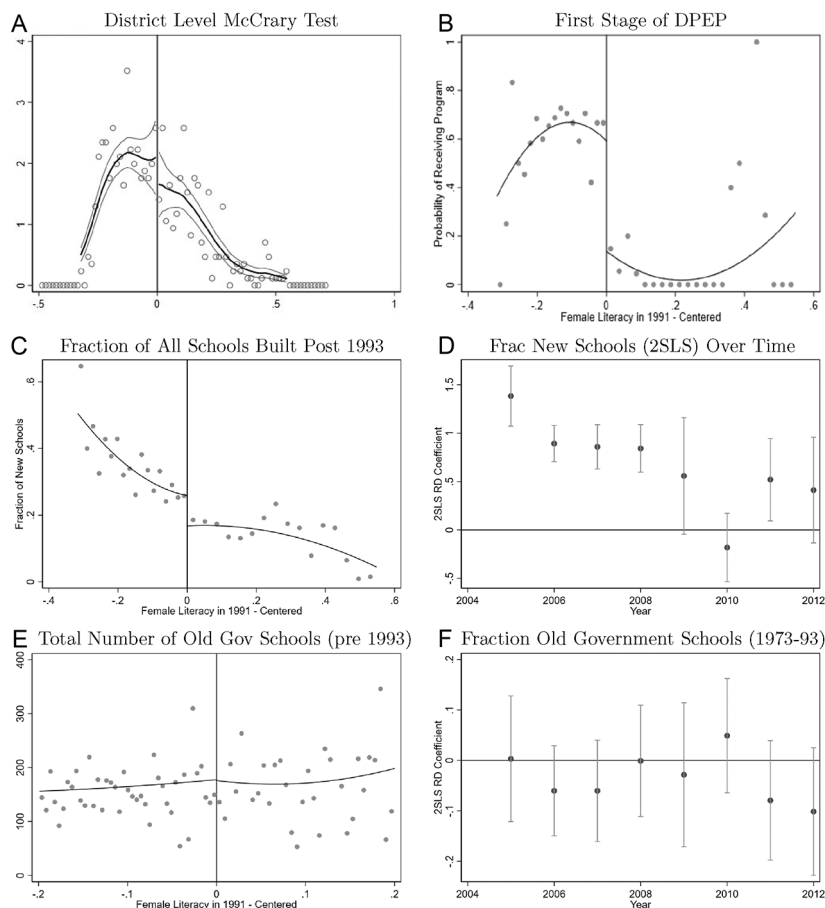


FIG. 1.—McCrary test, the first stage, and school building. *A*, McCrary (2008) test for discontinuity in density at the cutoff. *B*, First-stage graph, showing the probability that a district received DPEP funds, using Calonico et al. (2017). *C–F* use the DISE data; scatter plots (*C*, *E*) use the 2005 data. “New schools” are schools built after 1993. “Old schools” are schools built between 1973 and 1993. RD graph optimal binning and 2SLS (scaled up by the probability of treatment) RD coefficients were calculated with the Calonico, Cattaneo, and Titiunik (2014) procedure.

The first stage is presented in figure 1*B*. There is a sharp discontinuity at the cutoff, which provides a causal estimate of the local average treatment effect (LATE) for districts near the cutoff (Imbens and Angrist 1994). The parameters, like the estimated returns to education, are for students who were induced into more schooling and lived in districts near the cutoff that took up the policy. Similarly, the GE effects depend on what type of students get induced into more skill, as this may affect the amount of skill-biased capital adopted by the change in the effective

supply of labor. These GE effects, however, also affect subpopulations that were not induced into getting more education.

Estimating causal impacts requires that there is no perfect manipulation of the running variable or the cutoff, which is likely here, as the cutoff was the national average of the female literacy rate from the previous census. McCrary (2008) tests indicate no discontinuity in the density of districts around the cutoff (fig. 1A), and the p -value of the change in density is .71. Importantly, figure 1A also shows that there is a high density of districts at the national average, providing sufficient data for RD estimation. Other falsification tests discussed below justify the RD assumptions that there were no other discontinuities at the same cutoff.²²

While I present RD results graphically, the coefficients of interest are calculated with two different optimal bandwidth procedures, by Imbens and Kalyanaraman (2012) and Calonico, Cattaneo, and Titiunik (2014). Imbens and Kalyanaraman (2012) use a data-driven bandwidth selection algorithm to identify the optimal bandwidth for a local linear regression, given a squared loss function, whereas Calonico, Cattaneo, and Titiunik (2014) perform a bias correction and develop robust standard errors. Results using both bandwidth procedures are presented and are robust to using other parametric approaches (Hahn, Todd, and van der Klaauw 2001; Imbens and Lemieux 2008).²³

A. Using Policy Changes to Estimate Parameters

Variation in schooling, s_{id}^* , is driven by the variation in the marginal costs of schooling r_{id} . Since the costs of schooling are likely to be correlated with the ability of the worker ($\text{Cov}(\eta_i, \epsilon_i) \neq 0$) and there are underlying baseline differences in the skill distribution and skill-biased capital across these markets (eq. [4]), OLS regressions of earnings on education will be biased. In section II and appendix B.1, we derived that the equilibrium amount of aggregate schooling in a district, S_d^* , is affected by schooling expansions:²⁴

$$S_d^* = \phi_1 \mathbb{E}[\tilde{\beta}_{asd}] + \phi_2 R_d - \frac{\eta_d}{\Gamma}. \quad (10)$$

The $\phi_2 R_d$ portion captures how more government spending increases equilibrium schooling by making public schools more accessible and

²² Cattaneo, Jansson, and Ma (2017) offer an alternative test for manipulation at the cutoff that does not rely on the selection of binning parameters. The p -value of a discontinuity in the density using their method is .97.

²³ The results are robust to using various alternative procedures (table A.12). Bartalotti and Brummet (2017) allow for standard errors at an aggregated level. Calonico et al. (2017) allow for different-sized optimal bandwidths on either side of the cutoff and for nearest-neighbor standard errors.

²⁴ See app. B.1 for a parameterization of ϕ_1 and ϕ_2 , where $\phi_1 \equiv [\bar{\theta}_d^2 / (\Gamma \bar{\theta}_d^2 + z_{2d})] > 0$, $\phi_2 \equiv \{(z_{2d} + \Psi \bar{\theta}_d^2) [\prod_m (\alpha_m / p_m)^{\alpha_m}] / (\Gamma \bar{\theta}_d^2 + z_{2d})\} > 0$, and $\eta_d = \mathbb{E}[\eta_i | i \in d]$.

making (via adjustments in the market price) private schools more affordable (app. B.1). The term $\phi_1 \mathbb{E}[\tilde{\beta}_{asd}]$ captures how changes in expected returns to education affect equilibrium schooling. If, for instance, the labor market GE effects are expected to substantially lower the returns to education $\mathbb{E}[\tilde{\beta}_{asd}]$, then there may be no increase in the equilibrium amount of schooling. The final term, η_d/Γ (baseline residual schooling costs), is unaffected by schooling expansions. We would, however, expect it to be correlated with other unobserved district characteristics. Yet η_d are likely not different for districts that fall just on either side of the cut-off, allowing the RD to recover unbiased estimates.

Let us define $D_d = 1$ to be districts that fall just on the side of the cut-off that receives the policy and $D_d = 0$ to be districts that fall just on the other side. In the neighborhood of the cutoff, we should therefore expect, for $\phi D_d \equiv \phi_1 \mathbb{E}[\tilde{\beta}_{asd}] + \phi_2 R_d$,

$$S_d = \phi D_d - \frac{\eta_d}{\Gamma} \quad \text{and} \quad \mathbb{E}[\eta_d | D_d = 1] = \mathbb{E}[\eta_d | D_d = 0]. \quad (11)$$

If the direct effects of increasing access to schooling outweigh any negative labor market GE effects that depress returns, then we should expect $\phi > 0$.

B. Returns to Education and Disentangling Earnings

The model derives equations for the returns to education as a function of quantities of labor and skill-biased capital. This can be strictly linked to the policy, which changes the distribution of earnings across the RD cutoff. In equation (3), reproduced below,²⁵ ψ_a captures the cohort effect, θ_{sd} captures the pure productivity effect and changes in skill-biased capital, the term $(1/\sigma_A) \log \ell_{asd}$ is crucial for the cohort-specific labor market GE effect, and $(1/\sigma_E) \log Y_d + [(1/\sigma_A) - (1/\sigma_E)] \log L_{sd}$ determines the GE effect that affects all cohorts.²⁶

$$\begin{aligned} \log w_{asd} = & \log \tilde{q} + \log \theta_{sd} + \log \psi_a + \frac{1}{\sigma_E} \log Y_d \\ & + \left(\frac{1}{\sigma_A} - \frac{1}{\sigma_E} \right) \log L_{sd} - \frac{1}{\sigma_A} \log \ell_{asd}. \end{aligned} \quad (3)$$

²⁵ For clarity, I repeat the same equation number every time the same equation appears in the text.

²⁶ While this equation is represented in terms of production function parameters, the estimated GE effects will not depend on the specific functional form of the production function as long as workers can be disaggregated into skilled vs. unskilled and young vs. old. The functional form is to better understand the role played by underlying economic parameters.

I leverage variation along various dimensions (age cohorts, skill levels, and treatment status) to disentangle the components of the change in earnings across the RD cutoff. When comparisons are restricted to being within cohorts, the cohort effect on earnings Ψ_a is differenced out. Cohorts in treated districts that were too old to change their education at policy implementation will be partially affected by labor market GE effects. The GE effects that affect all cohorts can thus be isolated by looking at the impact on the skill premium of older cohorts.

Earnings for the young will additionally be affected by cohort-specific GE effects, as there are more skilled workers, particularly in younger cohorts. As the young and old are not perfect substitutes and may be complements in production, I estimate effects for each cohort separately. While the estimates are informative for the model, such natural experiments also provide well-identified evidence. To that end, for unbiased estimation, I follow a few rules similar to Jackson's (2021) analysis of difference-in-RD designs. First, I compare outcomes across the RD cutoff; second, I compare the same cohorts across the cutoff (to account for cohort effects); and last, I compare the same skill group across the cutoff.

For ease of exposition, I restrict the analysis to two skill levels—skilled (s) and unskilled (u) workers. The fractions of each among the young y are represented by ℓ_{sy} and ℓ_{uy} , respectively. For any two-skill groups, $\Delta\ell_{sy} \equiv (\ell_{sy,D=1} - \ell_{sy,D=0}) = -\Delta\ell_{uy} \equiv (\ell_{uy,D=1} - \ell_{uy,D=0})$.

If only a single individual was to acquire skill and change status from unskilled u to skilled s , the GE effects would be infinitesimally small. If the person lives in the untreated region $D = 0$, then that person's earnings on acquiring skill would increase by²⁷

$$\begin{aligned} \log \frac{w_{as,D=0}}{w_{au,D=0}} &= \underbrace{\log \frac{\theta_{s,D=0}}{\theta_{u,D=0}}}_{\text{Productivity}} + \underbrace{\left(\frac{1}{\sigma_A} - \frac{1}{\sigma_E} \right) \log \frac{L_{s,D=0}}{L_{u,D=0}}}_{\text{Aggregate skill distribution}} \\ &\quad - \underbrace{\frac{1}{\sigma_A} \log \frac{\ell_{as,D=0}}{\ell_{au,D=0}}}_{\text{Cohort-specific skill distribution}} \equiv \beta_{as,D=0}, \end{aligned} \quad (12)$$

where $\beta_{as,D=0}$ are the earnings returns to changing one's skill from u to s in district $D = 0$. If, however, the individual lived in a treated region $D = 1$, where there are a lot more educated people or skill-biased capital because of the policy, then the change in earnings would be

²⁷ There is no expectations operator, as these are the actual aggregate returns and not individual expectations.

$$\begin{aligned} \log \frac{w_{as,D=1}}{w_{au,D=1}} &= \log \frac{\theta_{s,D=1}}{\theta_{u,D=1}} + \left(\frac{1}{\sigma_A} - \frac{1}{\sigma_E} \right) \log \frac{L_{s,D=1}}{L_{u,D=1}} \\ &\quad - \frac{1}{\sigma_A} \log \frac{\ell_{as,D=1}}{\ell_{au,D=1}} \equiv \beta_{as,D=1}, \end{aligned} \quad (13)$$

where $\beta_{as,D=1}$ is defined as the earnings returns to changing ones skill from u to s in treated regions $D = 1$. These returns differ across regions because of the differences in the amount of skill-biased capital ($\log(\theta_{s,D=1}/\theta_{u,D=1}) - \log(\theta_{s,D=0}/\theta_{u,D=0})$), the size of the skilled workforce ($\log(L_{s,D=1}/L_{u,D=1}) - \log(L_{s,D=0}/L_{u,D=0})$), and also the size of the young skilled workforce ($\log(\ell_{as,D=1}/\ell_{au,D=1}) - \log(\ell_{as,D=0}/\ell_{au,D=0})$).

The difference in the returns to acquiring skill between these two regions is $\Delta\beta_{as} \equiv \beta_{as,D=1} - \beta_{as,D=0}$. Across the RD cutoff, these returns will be different because of a change in the skill composition of the workforce and the adoption of skill biased capital. The term $\Delta\beta_{as}$ captures the GE effects on the returns to skill:

$$\begin{aligned} \Delta\beta_{as} &= \underbrace{\left(\log \frac{\theta_{s,D=1}}{\theta_{u,D=1}} - \log \frac{\theta_{s,D=0}}{\theta_{u,D=0}} \right) + \left(\frac{1}{\sigma_A} - \frac{1}{\sigma_E} \right) \left(\log \frac{L_{s,D=1}}{L_{u,D=1}} - \log \frac{L_{s,D=0}}{L_{u,D=0}} \right)}_{\text{GE effects on all cohorts}} \\ &\quad - \underbrace{\frac{1}{\sigma_A} \left(\log \frac{\ell_{as,D=1}}{\ell_{au,D=1}} - \log \frac{\ell_{as,D=0}}{\ell_{au,D=0}} \right)}_{\text{Additional GE effect on young}}. \end{aligned} \quad (14)$$

To estimate returns and the GE effects, I will not need to estimate every economic parameter (such as σ_A and σ_E). Such an approach is similar to ones taken in public economics (Hotelling 1938; Harberger 1954, 1964), sometimes measuring GE effects (Goulder and Williams 2003). Similar to what Heckman and Vytlačil (2007) refer to as “Marschak’s Maxim” (Marschak 1953), I need to identify only a combination of economic parameters rather than every primitive. This is less demanding of the data, while allowing for clean identification.

In order to disentangle the GE effects by cohort, I look at the discontinuity in the skill premium of the younger and older cohorts separately. By restricting the population to a specific skill level (and cohort), one ensures that the differences in earnings across the RD cutoff are due only to differences in the skill distribution and the amount of skill-biased capital.

The change in returns in equation (14) can be split into two components. The first is the GE effect that affects all cohorts. To estimate this effect, I examine the change in the skill premium for the older cohort o .

Empirically, this is the earnings differential between the skilled older population and the unskilled older populations:²⁸

$$\underbrace{\log \frac{w_{so,D=1}}{w_{so,D=0}} - \log \frac{w_{uo,D=1}}{w_{uo,D=0}}}_{\text{GE effects on all cohorts}} = \underbrace{\left(\log \frac{\theta_{s,D=1}}{\theta_{u,D=1}} - \log \frac{\theta_{s,D=0}}{\theta_{u,D=0}} \right)}_{\text{Skill-biased capital}} + \underbrace{\left(\frac{1}{\sigma_A} - \frac{1}{\sigma_E} \right) \left(\log \frac{L_{s,D=1}}{L_{u,D=1}} - \log \frac{L_{s,D=0}}{L_{u,D=0}} \right)}_{\text{Aggregate skill distribution}}. \quad (15)$$

Note that we would expect that these two portions of the GE effects on all cohorts counteract each other. On the one hand, an increase in the skilled workforce leads to the adoption of skill-biased capital and raises the skill premium. On the other hand, increasing the relative supply of skilled workers makes them less valuable, lowering the skill premium.

The second component of the GE effects from equation (14) is the additional GE effect on the young y , driven solely by the change in the age-specific skill distribution. This component can be measured by estimating the earnings differential between the skilled young and unskilled young, and differencing out the earnings differential between the skilled unskilled old:²⁹

$$\underbrace{\left(\log \frac{w_{sy,D=1}}{w_{sy,D=0}} - \log \frac{w_{uy,D=1}}{w_{uy,D=0}} \right) - \left(\log \frac{w_{so,D=1}}{w_{so,D=0}} - \log \frac{w_{uo,D=1}}{w_{uo,D=0}} \right)}_{\text{Additional GE effect on young}} = - \frac{1}{\sigma_A} \underbrace{\left(\log \frac{\ell_{ys,D=1}}{\ell_{yu,D=1}} - \log \frac{\ell_{ys,D=0}}{\ell_{yu,D=0}} \right)}_{\text{Age-specific skill distribution}}. \quad (16)$$

Migration directly affects the quantities in equations (15) and (16). For instance, if there is differential migration and skilled workers migrate out of the treated districts in search of work, then migration will weaken the strength of the “aggregate skill distribution” component of the GE effects by altering the size of the skilled workforce in treated districts. In this way, the model incorporates migration in determining the GE effects. The individual decision to migrate and the corresponding labor supply curve are specifically modeled in appendix B.6.

²⁸ Regardless of the specific form of the production function, the change in the skill premium for the old will be the GE effects on all cohorts, and estimates of returns and cohort-specific GE effects will empirically rely on the left-hand sides of eqq. (15) and (16). For instance, this is true even if wage returns are determined in a purely signaling model. The right-hand sides merely help us understand the underlying economic parameters.

²⁹ Note that if $\sigma_A < \sigma_E$, then the two components may be of opposite signs.

To estimate the two different returns $\beta_{as,D=0}$ and $\beta_{as,D=1}$, I use discontinuities in the average earnings of the young and the wages of the skilled young and unskilled young:³⁰

$$\log \frac{w_{y,D=1}}{w_{y,D=0}} = \ell_{sy,D=1} \log \frac{w_{sy,D=1}}{w_{sy,D=0}} + \ell_{uy,D=1} \log \frac{w_{uy,D=1}}{w_{uy,D=0}} + \underbrace{\Delta \ell_{sy} \log \frac{w_{sy,D=0}}{w_{uy,D=0}}}_{\beta_{y,D=0}}, \quad (17)$$

$$\log \frac{w_{y,D=1}}{w_{y,D=0}} = \ell_{sy,D=0} \log \frac{w_{sy,D=1}}{w_{sy,D=0}} + \ell_{uy,D=0} \log \frac{w_{uy,D=1}}{w_{uy,D=0}} + \underbrace{\Delta \ell_{sy} \log \frac{w_{sy,D=1}}{w_{uy,D=1}}}_{\beta_{y,D=1}}. \quad (18)$$

The change in the average earnings for the younger cohorts is a weighted average of the change in the young skilled wage (weighted by the fraction skilled), the young unskilled wage (weighted by the fraction unskilled), and the returns to skill (weighted by the fraction of compliers). These relationships can be used to derive the returns to skill in the treated and untreated districts separately. At the same time, the average number of years of education in the districts changes across the cutoff in the following manner:

$$\begin{aligned} \Delta S &= (\ell_{sy,D=1} s_1 + \ell_{uy,D=1} s_0) - (\ell_{sy,D=0} s_1 + \ell_{uy,D=0} s_0) \\ &= \Delta \ell_{sy} s_1 + \Delta \ell_{uy} s_0 = \Delta \ell_{sy} (s_1 - s_0), \end{aligned} \quad (19)$$

where s_1 is the years of education for the skilled group, s_0 is the years for the unskilled group, and $\Delta \ell_{sy}$ is the fraction of students induced into getting more skill.³¹

C. Outcomes and Economic Benefits

The economic benefits depend on the changes in the wage distribution across the cutoff. Yet the benefits to different types of workers depend on a few crucial elasticities, as in Harberger (1964).³² For instance, the labor market benefits to workers induced into getting more skill are the sum of partial equilibrium returns and the GE effect on skilled wages. This highlights the importance of estimating both parameters in order to measure economic benefits.

The first determinant of the changes in overall benefits is the reduction in costs of schooling for younger cohorts. This is particularly meaningful

³⁰ See app. B.5 for detailed derivations of these equations.

³¹ Recall that the per-year rate of return is a product of the returns to skill and the education gap between the skilled and unskilled: $\beta_{asd} = \beta_{asd}/(s_1 - s_0)$.

³² Similar to what Heckman and Vytlačil (2007) refer to as “Marschak’s Maxim” (Marschak 1953), I need to identify only a combination of economic parameters rather than every primitive. This puts fewer constraints on the data by focusing identification on certain parameters.

for inframarginal students who were always going to attend school, even if the policy had not induced marginal students to get more education. Studies that focus on only the enrollment response in education interventions may miss this large component of benefits that affects inframarginal students.

Labor market benefits depend on the increase in overall output due to skill adoption and on the labor market returns. The increase in total output depends on the productivity parameters and the change in the skill distribution. At the same time, the GE effects will have distributional consequences. The welfare of older cohorts is unaffected by the reduction in the costs of schooling. The skilled old, however, are adversely affected by GE effects that affect all skilled workers, whereas the unskilled old benefit from increases in their earnings.

If agents do not have perfect foresight, we need to distinguish between welfare ex ante and ex post of GE effects.³³ Ex post welfare depends on $\beta_{as,D=0}$, the actual returns to skill in untreated districts, and $\beta_{as,D=1}$, the returns including the GE effects. The ex post welfare for a young high-skill person who would acquire skill even in the absence of the policy rises by the reduction in the total costs of education but is dampened by GE effects that affect all cohorts and additional GE effects on the young. Labor market welfare for them is $\log(w_{as,D=1}/w_{as,D=0})$. In the absence of GE effects, the young always-skilled are affected only by reductions in education costs. For those who would never acquire skill, even in the presence of the policy, the difference in the unskilled wage at the cutoff is their ex post labor market welfare: $\log(w_{au,D=1}/w_{au,D=0})$.

For younger cohorts, who are induced into getting more skill, the ex post labor market welfare change depends on the skilled wage in treated districts and the unskilled wage in untreated districts, $\log(w_{ys,D=1}/w_{yu,D=0})$, and thereby the returns $\beta_{ys,D=0}$:

$$\log \frac{w_{ys,D=1}}{w_{yu,D=0}} = \log \frac{w_{ys,D=0}}{w_{yu,D=0}} + \log \frac{w_{ys,D=1}}{w_{ys,D=0}} = \beta_{ys,D=0} + \log \frac{w_{ys,D=1}}{w_{ys,D=0}}. \quad (20)$$

Consequently, changes in ex post labor market benefits for those induced into getting more skill consist of two components: the partial equilibrium returns and the GE change in economic benefits to the “always-skilled.” This is why it is important to estimate both the partial equilibrium returns and the GE effects. In the absence of any GE effects, the change in earnings for a person induced into getting more education would simply be $\beta_{ys,D=0}$.

Ex ante welfare depends on expectations. If expectations were myopic, then wage expectations in regions that receive the policy, $D = 1$,

³³ I measure labor and education market “welfare” in log monetary units, as they depend on costs and earnings.

would equal steady-state values in the absence of the policy (i.e., wages in untreated regions, $D = 0$). Consequently, $\mathbb{E}[\beta_{as,D=1}] = \beta_{as,D=0}$. This ex ante myopic wage return is the partial equilibrium ex post returns that I estimate. Ex ante change in welfare would be similar to the partial equilibrium ex post welfare mentioned above.³⁴

If agents had perfect foresight, then ex ante welfare changes would be the same as the ex post GE changes above. If expectations were neither myopic nor of perfect foresight, then ex ante welfare depends on how wages are expected to change. For the young always-skilled, this is the expected change in skilled wages, and for the young never-skilled, this is expected changes in unskilled wages. For those induced into skill, welfare depends on how both skilled and unskilled wages are expected to change. I measure ex post welfare changes for different groups (by cohort and skill) and ex ante changes under perfect foresight and myopia.

To compare labor market gains to reduced schooling costs and estimate welfare changes, I discount labor market gains by the real interest rate over the period. For a student induced into more education, the costs include tuition and the opportunity cost of forgone unskilled wages. The benefits include the present discounted value of a skilled worker's earnings stream.

V. Results

A. Public and Private School Building

The primary objective of the program was to build new schools. Figure 1C shows the effect of the program on schools built once the program was underway in 1994. I trace out the longer-term effects by studying how the coefficient in figure 1D changes over time. The first coefficient, plotted for the year 2005, shows a large discontinuity in the fraction of new schools, whereas other coefficients, in later years, show a smaller difference among districts on either side of the cutoff, as funding declined. In the absence of funds, regions on the untreated side of the cutoff catch up over time by building schools at a relatively more rapid rate. As a falsification test, in figures 1E and 1F, I show no differential impacts on the fraction of schools that were built in the 20-year period before the program (1973–93).³⁵

How private schools respond to such interventions is crucial for determining the overall benefits. An expansion in public schooling may lower the competitive price that private schools can charge and price out the

³⁴ For the young always-skilled, myopic ex ante welfare change would just be the reduction in schooling costs. The young “never-skilled” see no change in myopic ex ante welfare. For the young induced into skill, the myopic ex ante labor market welfare change is simply the partial equilibrium returns $\beta_{ys,D=0}$.

³⁵ In fig. A.3, I split the sample by government and private schools and show alternative versions where I plot the total schools per capita and the dynamic trends for old private schools.

less efficient private schools. However, it is also possible for them to enter, given the likelihood of peer effects, self-segregation motives, and changes to the local economy and infrastructure driven by such a large-scale program. In figure A.3*b*, there is no evidence of crowd-out, and if anything, there is mild evidence of crowd-in.³⁶

School quality.—While the primary focus of DPEP was to increase educational attainment by building schools, there may have been quality improvements, given the large funding and hiring of teachers. In table A.3, I use the ASER data: of six different test score variables, only one (the ability to identify numbers) shows a statistically significant 5 percentage point increase. This is, at best, weak evidence of better test scores that may attenuate the negative GE effects. Either way, there is no detectable evidence on declining quality. On the other hand, better “quality” in terms of better infrastructure may have made it easier for students to finish a grade and so further lower the marginal costs of schooling.

B. Education and Earnings

Since DPEP was introduced in 1994, we would not expect changes in education for students who were past school-going age at the time. In appendix F.1, I discuss how cohorts are defined.³⁷ I find no detectable discontinuities in education, literacy, or different levels of education for older individuals in the left-hand panels of figures 2 and A.4 and the top two panels of tables 1 and A.6.

Table 1 and figure 2 show RD impacts on education and earnings for those who reported earnings, across different bandwidths and age groups. The ITT (intent-to-treat) estimates on the young are 0.7 more years of education and a 0.11 log points increase in earnings. The estimates for the full sample in table A.4 show that the young attain 0.217 more years of schooling. Treatment-on-the-treated (TOT) estimates, scaled up by treatment probabilities, are in table A.6. Older populations have no discontinuity in education and earnings. The old could still have GE effects, as average earnings conflate falls in skilled and rises in unskilled wages.³⁸

³⁶ What drives the crowd-in? On the one hand, the demand externality could raise the equilibrium tuition and draw in private schools; on the other, a fall in operating costs may induce private school entry and lower the equilibrium tuition. In app. B.1.2, I discuss how I determine which mechanism is stronger by seeing how tuitions changes. Later, in sec. V.D—specifically, table A.14—I show that household expenditure on schooling falls, suggesting that the cost reduction mechanism is stronger.

³⁷ As I show, results are robust to alternative age cutoffs. I show appendix tables with multiple age groupings and wider age restrictions. In DID specifications, I show impacts on each age separately.

³⁸ I find some negative effects on average wages for close substitutes: cohorts close to treated cohorts. In table A.7, the sample is sliced thin into more age groups, and even though they are imprecisely estimated, there do seem to be negative earning effects on 36–45-year-olds, the age group closest to the treated cohorts.

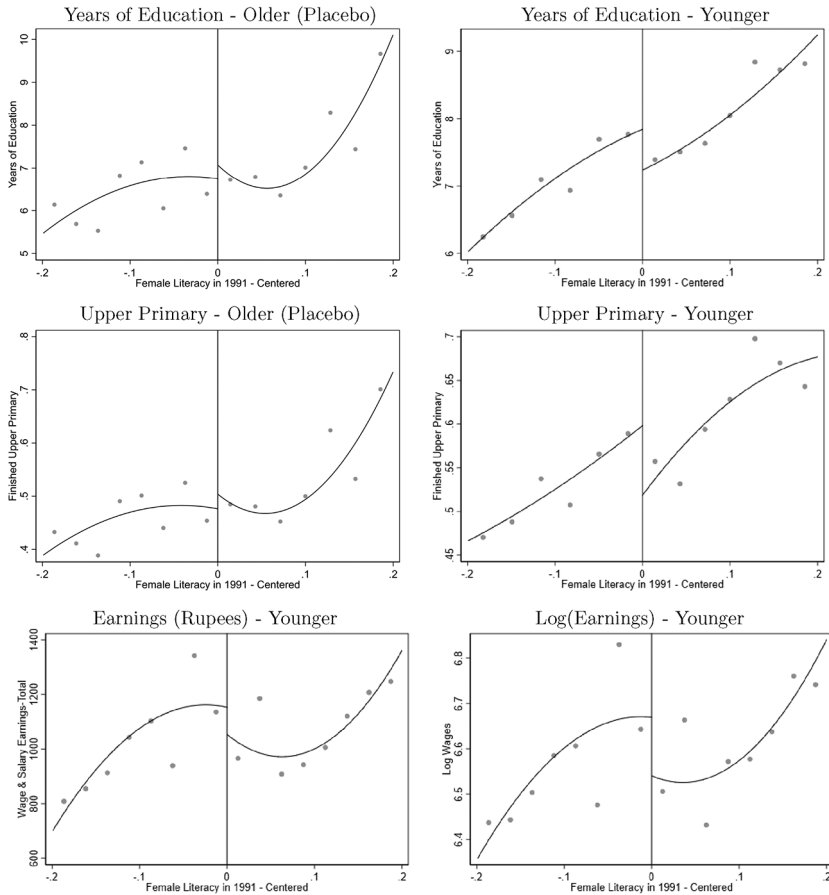


FIG. 2.—RD years of education and earnings. Source is NSS 2009 for those who reported earnings. Figures were made with the Calonico, Cattaneo, and Titiunik (2014) method of using regression curves to approximate the conditional means on either side of the cutoffs and the equally spaced sample means in optimally spaced bins. Average exchange rate in 2009 is 40 rupees = \$1.

The program was targeted toward primary and upper primary levels, and we expect the largest impacts at those levels. Table A.5 and the right-hand panels of figure A.4 show discontinuities in different levels of education for the young. Literacy rates are higher by 3 percentage points and the likelihood of finishing primary school by 5.8 percentage points. Below, I define “skilled” to be those who finished upper primary, as that is where the program was targeted, where the largest effects are, and where the sample is roughly divided into equal halves. Table A.4 and figure A.5

TABLE 1
EDUCATION AND EARNINGS FOR THOSE WITH REPORTED EARNINGS

	BANDWIDTH SELECTION: CCT METHOD		BANDWIDTH SELECTION: I AND K METHOD	
	Young	Old	Young	Old
A. First Stage Years of Education				
RD estimate	.720*** (.199)	−.0856 (.218)	.698*** (.173)	.100 (.188)
Finished Upper Primary				
RD estimate	.0743*** (.0197)	−.0169 (.0184)	.0733*** (.0169)	.000212 (.0158)
Observations	10,175	11,293	14,277	16,007
B. Reduced Form Log Earnings				
RD estimate	.112*** (.0312)	−.0114 (.0372)	.145*** (.0269)	.0432 (.0318)
Observations	10,175	11,293	14,277	16,007
C. 2SLS IV-LATE Conventional Method Returns Log Earnings				
Years of education	.155*** (.0427)	.129 (.303)	.208*** (.0460)	.442 (.666)
Observations	10,175	11,290	14,277	16,004

NOTE.—Data from NSS 2009–10, for all districts and all persons between the ages of 16 and 75 who reported earnings. The sample of “Young” are of school-going age during the policy, whereas those “Old” are too old to change their schooling in response to the policy. Bandwidths: “CCT” indicates the Calonico, Cattaneo, and Titiunik (2014) method, “I and K” the Imbens and Kalyanaraman (2012) method. Panel A report ITT effects. Panel C shows 2SLS regressions that treat the first-stage outcome as “years of education.” This is the ratio of “log earnings” from panel B and “years of education” from panel A. This approach is similar to conventional IV-LATE methods of computing the returns to education.

*** Significant at the 1% level.

show analogous results for the full sample, rather than for those reporting earnings.³⁹

Heterogeneity by gender, and robustness.—Since men may be likely to be in occupations that benefit from education, while women may be more likely to be engaged in domestic work, we may expect men to be more responsive to these interventions (Kingdon 1998; Drèze and Sen 2002). In tables A.9 and A.10, I find that the effects are concentrated among males, which is

³⁹ Comparing table 1 and figs. 2 and A.4, it is clear that, as in Duflo (2001), the impact on education is higher for the subsample that reported earnings. Yet, as the top panel of table A.8 shows, there is no discontinuity in the probability that earnings are reported at the cutoff, suggesting that DPEP did not lead to differential selection into who reported their earnings. In the NSS, the probability that earnings are reported is uncorrelated with working in agriculture or being self-employed. The difference in the educational impacts between those who reported earnings and the full sample can be tied to the difference in labor market returns by gender.

similar to the related literature (Breierova and Duflo 2004; Jalan and Glinskaya 2013; Ashraf et al. 2020). In the full sample, men increase their education by about 0.3 years, whereas women increase theirs by only about 0.09 years. For the subsample of those who report earnings, however, the impact on education is similar in magnitude but more precisely estimated for men. There is also little to no change in the earnings of women, even though men's earnings do rise.⁴⁰

In the appendix, I conduct a number of robustness checks. I collapse all the household data into district-age cells and rerun the regressions. Even as collapsing the data loses valuable information used in estimating the optimal bandwidth, the results do not change (table A.11). I try more in-progress RD bandwidth selection procedures and standard error estimation methods in table A.12, including methods that allow for different bandwidths on either side of the cutoff and nearest-neighbor variance estimation at district clusters. I test the sensitivity to age cutoffs by binning age groups into finer categories in table A.7 and the sensitivity to age restrictions by including a larger sample of ages in table A.13.

C. Returns to Education

1. OLS and Conventional IV Methods

In my sample, a simple OLS regression of log earnings on years of education and a quadratic age profile yields a Mincerian “return” of 10%. The IV estimates will estimate a 2SLS (two-stage least squares) LATE weighted by the probability of being induced into getting more education by the instrument. In general, IV-LATE estimates are found to be larger, perhaps as a reduction in marginal costs that affects all students equally will induce those with higher returns into getting more education (Imbens and Angrist 1994; Oreopoulos 2006; Carneiro, Heckman, and Vytlacil 2011).

Canonical IV-Wald methods estimate returns to education by using the RD cutoff to estimate first the change in education and then the corresponding change in earnings for the same cohort. By taking the ratio of the change in log earnings to the change in education, one estimates the returns to schooling. Under the assumption that the policy induces only some young workers to get more education, this method identifies the change in earnings due to an additional year of schooling for this marginal group. Yet as my model stresses, the policy simultaneously affects both the skill premium and the overall output in the district. Since changes in average earnings are driven not just by the switch in the fraction of students from unskilled to skilled groups but also by the changes in skilled and unskilled wages, the estimated individual returns are conflated with changes in output and the skill premium.

⁴⁰ One can add an additional nest to the CES production function that captures differences across genders.

The estimates in table 1 can be used to calculate the returns using the conventional method of taking the ratio of the change in log earnings and the change in years of education. The ratio of 0.112 log earnings and 0.72 years gives us a return of about 15.5%. Panel C of table 1 shows the 2SLS-LATE version of this exercise. This estimate is not statistically indistinguishable from numbers as low as 7% and lies within the range of comparable estimates found in the literature (Psacharopoulos and Patrinos 2004; Banerjee and Duflo 2005).⁴¹

The 2SLS return of 15.5% using the IV method described above ignores GE effects and so is neither the partial equilibrium return nor the return with the GE effects. As I show at the start of section V.C.3 below, it is a weighted average of both and lies between them.

2. DID Designs

The DID designs compare students across two dimensions: (1) whether the region received the policy and (2) whether cohorts were young enough to change schooling. As I show in appendix E.1, it is challenging to recover the partial (or GE) returns to schooling using an estimator that relies on “what happens to the old.” I derive what a DID analysis estimates and the assumptions needed to derive meaningful parameters from DID analyses.

There are at least two challenges with a DID estimator. The first is the “composition effect.” The young and the old differ in the fraction of skilled workers. As a result of the GE effects, skilled wages fall and unskilled wages rise. If there are more unskilled older workers, then the average wage for the old may actually rise. Consequently, the old may not be a good counterfactual for the young, as skill compositions differ by cohort. The second reason is a “complementarity effect.” The young and old are not perfect substitutes in production. More skilled young workers affect the wages of skilled older workers in different ways, given the cohort elasticity of substitution σ_A . If they are complements, then the wage for skilled older workers may actually rise. Again, the old may not be good counterfactuals for the young.

In table 2 and figure A.6, I compare DPEP districts to non-DPEP districts and the older cohorts to the younger cohorts.⁴² I estimate the DID

⁴¹ Recent experimental estimates of returns in a developing country are 13% (Duflo, Dupas, and Kremer 2021). While the bottom panel of table 1 shows log earnings for the old, as there is no meaningful first-stage response (in education), we should not interpret these as identified earnings returns.

⁴² For person i in age cohort a and district d , the following DID regression was estimated:

$$y_{ida} = \beta_{\text{DID}} T_{da} + \mu_d + \varpi_a + \epsilon_{ida}, \quad (21)$$

where μ_d is a district fixed effect, ϖ_a is a cohort fixed effect, and $T_{da} = 1$ if the individual lives in a DPEP district and is young enough to be affected. Under the parallel-trends assumption, β_{DID} is the DID parameter.

TABLE 2
DID (Full Model)

	(1)	(2)	(3)	(4)
Full Sample				
	Years of Education	Literate	Finished Primary	Finished Upper Primary
Estimate	.332*** (.0927)	.0551*** (.00783)	.0386*** (.00754)	.0196** (.00898)
Observations	279,452	279,483	279,483	279,483
R ²	.176	.189	.193	.170
Small Bandwidth				
	Years of Education	Literate	Finished Primary	Finished Upper Primary
Estimate	.311*** (.106)	.0426*** (.00764)	.0302*** (.00834)	.0209** (.00959)
Observations	144,248	144,261	144,261	144,261
R ²	.108	.118	.117	.103
Reported Earnings				
	Years of Education	Literate	Finished Primary	Finished Upper Primary
Estimate	.377** (.155)	.0558*** (.0111)	.0410*** (.0119)	.0299** (.0150)
Observations	66,093	66,098	66,098	66,098
R ²	.157	.166	.164	.139
	Log(Earnings)		2SLS Returns	
Estimate	.0596** (.0251)		.159*** (.0473)	
Observations	66,086		66,081	
R ²	.241		.201	
	Log (Earnings), Skilled	Log (Earnings), Unskilled	Additional GE on Young	
Estimate	-.0611** (.0281)	.0183 (.0211)	-.0794** (.0320)	
Observations	37,748	28,334		
R ²	.311	.225		

NOTE.—Data from NSS 2009–10, 17–75-year-olds. Regressions include district and cohort fixed effects. The DID coefficient is on the interaction between being below 35 and in DPEP district. Robust standard errors at the district level are in parentheses. “Small Bandwidth” restricts the sample in two ways: (1) ages are ± 15 years of the 35-year cutoff and (2) districts have female literacy $\in (-0.2, 0.2)$. “2SLS Returns” estimates two-stage least squares returns where the first-stage dependent variable is the years of education and the second-stage dependent variable is log earnings. “Additional GE on Young” estimates the GE effect that affects only the skill premium of the young (note: this excludes the average change in wages due to changes in output and the portion of the change in the skill premium experienced by all cohorts).

** Significant at the 5% level.

*** Significant at the 1% level.

coefficient for three different subsamples. For the full sample, there is an increase of 0.3 years in education and a 5.5 percentage point increase in literacy rates. There is also a 3.8 percentage point increase in the likelihood of finishing primary school. The estimates are similar even when the sample is restricted to be in the neighborhood of the RD cutoff and the cohort cutoff. For the subsample that reported earnings, there is also a statistically significant increase in earnings. The 2SLS IV-LATE “returns” can be estimated, taking the ratio of the change in log earnings and years of education. This DID-2SLS “return” is 15.9%. It is also possible to measure how the skill premium changes differentially for younger and older cohorts (an additional impact on the young). This component depresses the returns by 7.9 percentage points (table 2).

3. Returns to Education and the Labor Market GE Effects

The model allows me to estimate meaningful equations to calculate the GE effects and highlights an important point: the conventional method of taking the ratio of the younger cohort’s change in earnings and years of education is confounded by the fact that earnings are affected by the GE effects in the local economy. Equation (17) describes these returns:

$$\underbrace{\log \frac{w_{y,D=1}}{w_{y,D=0}}}_{\text{Avg. earnings}} = \underbrace{\ell_{sy,D=1}}_{\text{Skilled}} \underbrace{\log \frac{w_{sy,D=1}}{w_{sy,D=0}}}_{\neq 0} + \underbrace{\ell_{uy,D=1}}_{\text{Unskilled}} \underbrace{\log \frac{w_{uy,D=1}}{w_{uy,D=0}}}_{\neq 0} + \underbrace{\Delta \ell_{sy}}_{\text{Compliers}} \underbrace{\log \frac{w_{sy,D=0}}{w_{uy,D=0}}}_{\beta_{ys,D=0}}. \quad (17)$$

If there are no GE effects, then neither the skilled nor the unskilled wage should change, and so for changes in partial equilibrium, $\log(w_{uy,D=1}/w_{uy,D=0}) = \log(w_{sy,D=1}/w_{sy,D=0}) = 0$. The average wage would rise only because more people go from earning a low-/unskilled wage to earning a high-skilled wage ($\Delta \ell_{sy} > 0$). The change in average earnings across the RD cutoff would then recover the returns to skill for the compliers $\Delta \ell_{sy}$, since, under these assumptions, $\beta_{ys,D=0} = \log(w_{y,D=1}/w_{y,D=0})/\Delta \ell_{sy}$.

Sometimes referred to as the LATE theorem, this is often used to estimate the 2SLS-Wald returns to education, as I do in section V.C.1. Yet when there are GE effects, $\log(w_{uy,D=1}/w_{uy,D=0}) \neq 0$ and $\log(w_{sy,D=1}/w_{sy,D=0}) \neq 0$, and these confound the estimates. However, $\log(w_{uy,D=1}/w_{uy,D=0})$ and $\log(w_{sy,D=1}/w_{sy,D=0})$ are measurable quantities, and so all components of equation (17) are estimatable, allowing me to recover $\beta_{ys,D=0}$. Ignoring the GE effects produces an estimate (as in the bottom panel of table 1) that lies between the partial and general equilibrium returns to skill.

Average earnings of all persons in treated districts are affected by changes in overall output. At the same time, the change in skill distributions and the adoption of skill-biased capital affect skill groups and cohorts differently, as

TABLE 3
RETURNS AND WAGE PARAMETERS

	Fraction Switched	Change in Returns $\Delta\tilde{\beta}$	
Estimate	.171	−.066	
Standard error	(.045)	(.030)	
	Returns without GE ($\tilde{\beta}_{y,D=0}$)	Returns with GE ($\tilde{\beta}_{y,D=1}$)	Change in Returns (%)
Estimate	.199	.134	−33
Bootstrapped p -value	[.00]	[.00]	
	Change for Older Cohorts	Additional on Young	Change on Young (%)
$\Delta\tilde{\beta}$	−.0097	−.057	85.3

NOTE.—Data from NSS 2009–10. The estimation follows the procedures described in sec. IV.B and detailed in app. B.5, specifically eqq. (14), (17), and (18). The bootstrapping procedure and different components underlying the table estimates are described in detail in app. F.4. Younger cohorts are those between ages 17 and 35, whereas older cohorts are between ages 36 and 50. The p -values for returns with GE ($\tilde{\beta}_{y,D=1}$) and returns without GE ($\tilde{\beta}_{y,D=0}$) were bootstrapped using 1,500 draws of sampling with repetition. The null was created by jointly permutating the RD running variable, treatment status, and probability of treatment. The results in this table further suggest that the elasticity of substitution across age cohorts is approximately $\sigma_A = 5$, and in the absence of adoption of additional skill-biased capital the elasticity of substitution across skill groups would be $\sigma_E = 4.24$. Fraction switched is estimated in table A.6, third panel, first column.

in equation (14). While older cohorts are affected by the change in the aggregate skill distribution and inflow of skill-biased capital, younger cohorts are additionally affected by the change in the cohort-specific skill distribution for the young.

Given these GE effects, it is necessary to use the method outlined in section IV.B—and specifically, equations (17) and (18)—to derive the returns to education with and without the labor market GE effects. Scaling up by the treatment probability, there was a 17 percentage point increase in skilled workers across the cutoff (third panel of table A.6).⁴³

As returns depend on shares of skilled and unskilled (eq. [17]), in table 3 I bootstrap 1,500 draws with replacement, creating a null, jointly permutating the running variable, treatment status, and treatment probability. Appendix F.4 discusses the details. Estimated returns in the absence of GE effects are 19.9% per year (table 3). The returns with GE effects, however, are only 13.4%, a 33% decrease in returns attributable to GE effects.

⁴³ To reiterate, I define skilled workers as those finishing upper primary school, as the policy targeted this level and because the largest earnings increase in OLS regressions on untreated districts comes at finishing upper primary school. In going from literate-below primary to finishing primary school, average earnings increase by 10%, whereas in going from primary to upper primary, average earnings increase by 20%.

This change in the skill premium can be split into the portion that affects all cohorts and additional impacts only on the young. To do this, I use equations (15) and (16). Table 3 implies that 85.3% of the change in GE effects is from the “additional impact on young” term. The GE effect on all cohorts may be small, as the two components that determine this effect counteract each other—an increase in the relative supply of skilled workers ($\log(L_{s,D=1}/L_{u,D=1})$) lowers the skill premium, but adoption of skill-biased capital ($\log(\theta_{s,D=1}/\theta_{u,D=1})$) increases the premium.

Furthermore, the additional impact on the young is high, implying that the young and old are not close substitutes in production. Looking only at the GE effects on older cohorts considerably understates the GE effects on the young, and my paper is among the first to estimate the GE effects on all cohorts. Even as estimating the GE effects does not depend on the specifics of the model, they allow us to recover reliable elasticities. The elasticity of substitution across age groups $\sigma_A = 5$ is similar to that in Card and Lemieux (2001). In the absence of the adoption of skill-biased capital, the elasticity of substitutions across education groups would be $\sigma_E = 4.24$, yet the differential adoption of skill-biased capital inflates this figure.

4. External Validity, Compliance, and Extrapolating away from the Cutoff

There are two issues related to external validity. The first is related to the difference between complier and noncomplier districts (the RD is fuzzy), and the second is about extrapolating away from the RD cutoff. Appendix E.2 has a comprehensive discussion of these issues. The tests, with caution, lend support to the external validity of my estimates.

First, appendix E.2.1 tests for differences in outcomes between treated complier districts and always-takers and between untreated compliers and never-takers. Following Bertanha and Imbens (2020), in figure E.1 and table E.1, I test for discontinuities in education (*a*) conditioning on receiving the program and (*b*) conditioning on not receiving it. Then, in appendix E.2.2, I estimate the marginal treatment effect (MTE) across the distribution of the unobserved net costs of being treated and find that the MTE is relatively stable, suggesting a lack of meaningful treatment effect heterogeneity (Brinch, Mogstad, and Wiswall 2017). I also test for the size of the first stage by different district characteristics, in appendix E.2.3, and fail to detect meaningful differences in compliance probability across different occupations and industries, suggesting that baseline economic characteristics are unlikely to drive compliance probabilities.

Second, I discuss effects away from the RD cutoff. The advantage of my context is that the Indian government chose the cutoff to be the national average of female literacy, and so my estimates are already relevant for an attractively representative set of districts. In appendix E.2.4, I closely

follow Dong and Lewbel (2015), who estimate the marginal threshold treatment effect (MTTE), or how the treatment effect changes from a marginal change in the RD cutoff. The results in table E.3 show that, at least locally, the MTTE is statistically indistinguishable from zero and economically small. These results suggest that locally extrapolating around the cutoff would produce minor changes to the estimated treatment effects. Yet we should be cautious in extrapolating this to extremes of the running variable.

5. Returns Vary across Levels of Schooling

Following Lochner and Moretti (2015), in appendix E.3 I examine returns across different levels of education. The OLS estimates in figure E.4 suggest that, other than at the college level, there are no substantial nonlinearities across different levels of education. Montenegro and Patrinos (2014) find that in India, the annual OLS returns are higher for college graduates (20.8%) than for primary (5.8%) and secondary (6%) students (who have similar returns). Figure E.4 reflects this but makes another important point—the 2SLS weights are zero on college but are high on finishing middle (upper primary school), perhaps because the policy was targeted and mainly induces children to finish upper primary school. Consequently, the returns we estimate are not capturing returns to college.

6. Labor Market Distortions

How do labor market frictions affect the analysis at hand? Let us consider two types of frictions described in the literature: (i) misallocation of factor inputs across sectors (or firms) and (ii) monopsonistic labor markets (Muralidharan and Niehaus 2017). In appendix B.7, I discuss in detail the consequences of the former distortion. If there are J sectors, with factor misallocation, then the wages are no longer the same across sectors. I show that much of the analysis is unaffected and derive the conditions under which the estimated returns are simply a weighted average of sector-specific returns under certain conditions. In appendix B.8, I examine the consequences of market power in the labor market. While market power does not affect the private returns to workers, it will affect the impacts on productivity and output. To elaborate, the private returns to workers will continue to depend on the skill premium, but the difference between the skilled and unskilled wages no longer captures the difference between skilled and unskilled productivity. I derive an adjustment term that depends on the relative (between skilled and unskilled) labor supply elasticities, allowing us to identify the productivity consequences of upskilling, given known elasticities. As some recent development literature suggests

little difference in labor supply elasticities across subgroups (Goldberg 2016), this adjustment term is likely to be small.

*D. Total Output, Consumption,
and Educational Expenditure*

Changes in overall output depend on the productivity of different skill levels and the shift in the labor force across skill levels. As workers acquire skill and/or if skill-biased capital is adopted, overall productivity and output in the region increase. The top panel of figure A.7 shows the impact on the district domestic product. These regressions are underpowered and their standard errors large. The (imprecise) point estimates indicate that in 2000–2004, the increase in GDP (gross domestic product) associated with the policy was between 0.068 and 0.11 log points (table A.15).

Changes in total output lead to changes in total consumption. The top panel of table A.14 shows that the change in consumption expenditure in the last year of the policy (2004–5) was about 0.06 log points. At the same time, in 2004, the money spent for educational purposes (tuition, fees, books, and stationery) fell by between 0.085 and 0.21 log points.⁴⁴ This fall in educational expenditure is persistent even 5 years after the program ended, driven by lower spending on tuition and fees (bottom panel of figure A.7). There is, in fact, an increase in complementary expenditures, such as books and stationery (table A.14).

Changes in consumption and the costs of education will directly affect overall economic benefits. The increase in output and consumption benefits all cohorts, whereas the fall in the costs of education benefits younger cohorts who attend school. The fall in the costs of schooling even benefits inframarginal households, who would always get education, even in the absence of the policy, and even as they may not change their enrollment decisions. At an extreme, policies that successfully reduce the costs of schooling can have significant economic benefits for the inframarginal students even as they do not change the years of education.

E. Migration

Local economies receiving educational funds for at least a decade witnessed a transition in the skill level for younger cohorts in their workforce. For this to have happened, any combination of the following four things may have taken place. First, skilled workers may have migrated out, and this migration would dampen any GE effects that depend on the skill distribution. Second, existing firms may have switched the composition of their

⁴⁴ Jalan and Glinskaya (2013) measure a 20%–40% fall in household educational expenditure.

workforce by hiring more skilled workers. Third, new firms may enter and hire these skilled workers. Last, workers may utilize their increase in skill and adopt new technologies in production. The adoption of skill-biased capital, therefore, increases the returns to skill and affects the GE effects.

The migration model extension in appendix B.6 explicitly allows migration to change quantities of labor in equation (14). Worker mobility tends to equalize wages across regions and mitigate negative GE effects on the skilled or positive GE effects on the unskilled. This would attenuate the GE effects, as skilled workers would leave once their wages fell.⁴⁵ Yet permanent worker migration was extremely low in India, making it unlikely that newly skilled workers emigrate from these districts (Munshi and Rosenzweig 2009, 2016; Topalova 2010).⁴⁶

By analyzing the NSS 2007–8 waves, which asks questions on migration, I find that of all the households that reported having any migrants across districts, only 30% of the migration was work related, whereas more than half were for marriage. The migration model extension in appendix B.6 also derives an intuitive result: the change in migration rates depend on migration probabilities conditional on skill. Since empirically, $P(\text{migrate}|\text{skilled}) - P(\text{migrate}|\text{unskilled}) \approx 0$ and low for work-related migrants, we may not expect to see a migration response. Panel B of table A.8 shows that the policy did not meaningfully affect either any migration or work-related migration, for either the young or old cohorts.

F. Productivity, Capital Adoption, and District Heterogeneity

Local enterprises may adopt newer technologies or capital in response to the policy (Ghani, Goswami, and Kerr 2015), as I discuss in appendix F.3. The empirical results using the ASI data are in figure F.1. Even at the establishment level, average wages paid to workers increase as more educated workers start joining the labor market around 2004. Furthermore, I classify firms on the basis of their products as “high-skill” firms. The figure shows a steady increase in the fraction of firms that produce more mechanized products. This suggests that either existing firms shifted production and employed high-skill workers or newer firms entered and

⁴⁵ Regions around the RD cutoff are geographically dispersed, making the migration of capital or workers across cutoff regions less likely. Indeed, the little flows that occur were to big cities and not other poor areas.

⁴⁶ Many studies on India ignore migration, as the numbers are low (Das Gupta 1987; Foster and Rosenzweig 1996; Anderson 2005; Duflo and Pande 2007; Banerjee, Iyer, and Somanathan 2008). Munshi and Rosenzweig (2009) show that cross-village migration rates were only 8.7%, most of which perhaps takes place within the district. Deshingkar and Anderson (2004) show that rural-urban migration is much lower in India than other countries. Munshi and Rosenzweig (2016) show that worker migration is extremely low, despite large regional wage gaps.

hired these skilled workers. Both are suggestive of the adoption of skill-biased capital in these regions.

In general, there are some clear changes to the labor market for the workers in these regions. The bottom half of table A.8 shows that the probability of being paid monthly (as opposed to daily) is higher, and the fraction unemployed is lower in the treated regions. The last possibility—that workers adopted newer technologies, given their increased levels of education—is therefore possible in this context (Foster and Rosenzweig 1996).

In addition, it is important to note that the returns are not an immutable parameter but rather an endogenous outcome tied to local skill composition, productivity, and capital. In appendix F.2, I discuss heterogeneity in returns across district and labor market characteristics, such as the share employed in different sectors and strength of the labor market.

G. Overall Economic Benefits

The program raised aggregate economic benefits but had meaningful distributional consequences. Increases in overall output and reductions in the total cost of schooling will benefit households. The change in labor market earnings depends on the returns to skill and the GE effects on these returns. Table 3 shows the returns by skill group, which helps back out the parameters and the changes in yearly labor market benefits shown in table 4. These estimates depend on the TOT effects on earnings, scaled up by the probability of treatment. For these calculations, I use the average real interest rate during that period (5%). A gap of 10 years is assumed between when the costs of education are borne and labor market returns are realized. Finally, I distinguish between welfare *ex post* and *ex ante* of GE changes, as in section IV.C.⁴⁷

In the top panel of table 4, I present results for those in younger cohorts who were induced into getting more skill because of the policy. This is about 17% of the young population. Their *ex post* welfare (or *ex ante* perfect-foresight welfare) increases by 0.12 log points. The GE effects depress this increase in *ex post* welfare by 23.8%. Consequently, *ex post* welfare in partial equilibrium or *ex ante* myopic welfare would have increased by 0.157 log points.

At the same time, workers who were always going to acquire skill even in the absence of the policy are worse off by 0.037 log earnings points. This is their *ex post* GE (or *ex ante* perfect-foresight) labor market welfare change. For workers who were always going to be unskilled, their *ex post* labor market welfare (which is also their *ex ante* perfect-foresight welfare) rises by 0.014 log earnings points. *Ex post* welfare in partial equilibrium (or *ex ante*

⁴⁷ The average real interest rate comes from the World Bank World Development Indicators. Changing the interest rate or the gap of 10 years does not affect the percentage change in welfare due to the GE effects, only the levels.

TABLE 4
LABOR MARKET BENEFITS

	With GE	Without GE	Change (%)	Fraction of Cohort
Change in yearly labor market benefits for:				
Young induced into getting more skill	.120	.157	-23.8%	.17
Always skilled (young)	-.037	0	...	.39
Always unskilled (young)	.014	0	...	.44
	Among Old with GE	Among Old without GE	Among Young with GE	Among Young without GE
Transfer in yearly benefits from skilled to unskilled	.007	0	.052	0
	Costs	Benefits	Net	Change Due to GE (%)
Change in lifetime welfare for induced students	5.15	6.55	1.4	-23.8

NOTE.—Welfare numbers are in monetary log points. GE = general equilibrium effects. “Change in Benefits” is shown for the subpopulation that was young and changed their years of education to acquire skill. This is split into “with GE” effects and a possible counterfactual of what would happen to their welfare in the absence of GE effects (“without GE”). “Change” is defined as change in welfare; “without GE” is the base. “With GE” is ex post welfare when GE effects are included. It also tells us the ex ante welfare for those with perfect foresight and full information on returns. “Without GE” is ex post welfare if there were no GE effects. It also tells us ex ante welfare for those who have perfectly myopic expectations and know the wage distribution at the time they make schooling decisions. “Induced into getting more skill” indicates the population that switched from unskilled to skilled only because of the policy. “Always skilled” indicates the population that would have acquired skill even in the absence of the policy. “Always unskilled” indicates the fraction of the population that would not have acquired skill even in the presence of the policy. “Fraction switchers” is the estimated difference (across the RD cutoff) in subpopulations that acquired a higher level of education. Yearly welfare calculations assume an interest rate of 2.37% and a gap of 10 years between the costs of education and the labor market returns. Real interest rates are from the World Bank. The World Bank uses the lending rate and adjusts it for inflation, using the GDP deflator. For the period 2010–13, the average real interest rate was 2.37%. For “Change in lifetime welfare for induced students,” costs include (a) opportunity cost of forgone earnings for unskilled work and (b) tuition costs for students in DPEP districts near the cutoff. Costs are calculated in 2004 (NSS 61st round), and benefits include present discounted value of lifetime earnings stream assuming a real interest rate of 2.37%.

welfare for those with myopic expectations) would not change for these groups.⁴⁸

Since unskilled workers are better off and skilled workers worse off ex post, I estimate the transfer in labor market benefits from the skilled to the unskilled due to the GE effects. Among the old this transfer is

⁴⁸ Note that these results focus on labor market benefits. A policy such as this should also change the prices of nontradables, such as land, affecting the welfare of nonworkers as well. Given the scant number of land transactions in the data, there is no discernible effect on land prices.

0.007 log points, and among the young it is 0.05 log points. An analysis that ignores GE effects would overestimate the welfare on the young induced into skill by 23.8% and miss the transfer in welfare from the always-skilled to the never-skilled for each cohort.

To measure the lifetime welfare change for students induced into more schooling, I compare the cost of an additional year of schooling to the benefits in table 4. These costs include not just tuition fees but also the opportunity cost of forgone unskilled wages. The benefits are the present discounted value of the skilled earnings stream. Everyone benefits from increases in overall output, and the educated young benefit from reductions in schooling costs.

VI. Conclusion

Large-scale education investments can and do generate substantial GE effects in the labor market and the education sector. Bringing together a school-level data set, census data, household surveys, and firm-level data, I perform an intensive analysis of a schooling-expansion program that measurably increased educational inputs and years of education and earnings for students. Leveraging the policy, I estimate the parameters of a GE model using an RD design. My estimates imply that the per-year return to acquiring skill is 13.4%, or 6.6 percentage points lower than it would be in the absence of GE effects. These changes imply elasticities of substitution across skill groups and cohorts that are in line with previous literature in other contexts (Card and Lemieux 2001). There are also large distributional effects, where labor market benefits are transferred from the skilled to the unskilled, especially among the young. High-skill workers who would have acquired skill even in the absence of the policy lose out in terms of labor market earnings. Overall welfare, however, is higher, driven by decreases in the costs of education and increases in local economic output.⁴⁹

These findings have two important implications. First, the results in this paper help explain why scaled-up government policies may have impacts different from those of researcher-led microinterventions (Heckman, LaLonde, and Smith 1999; Acemoglu 2010; Deaton 2010). Identifying who benefits and who does not, and what works and what does not, is key to making such large-scale infrastructure investments more targeted and effective. The methods in this paper can be used to quantify the expected impacts of scaled-up microinterventions in other contexts. It is clear that understanding all the consequences of large GE effects is crucial

⁴⁹ These results do not necessarily indicate that the policy was cost-effective. I have shown that the direct impacts were concentrated on certain cohorts and had low persistence. Given the large amounts of funds invested, the overall cost-effectiveness of this policy is questionable and is left for future research.

for both researchers and policy makers when considering nationwide interventions in public policy.

Second, it speaks to the large body of literature using large-scale variation (tuition subsidies, compulsory-schooling laws, distance-based measures, and school building) to estimate the returns to education. Macro-level variation estimates a different parameter and may conflate the individual returns and GE effects. This is because an experiment where a single individual receives more education is inherently different from the variation induced by changes that affect entire cohorts of students.

While the methodology I develop is applicable to other similar settings across the world, my estimates are for local labor markets and not for the entire country. As such, they are not generalizable to regions farther away from the RD cutoff in the absence of stronger assumptions. Yet the great advantage of the RD cutoff is that it was for districts around the average female literacy; therefore, we should think of these results as pertaining to the average district.

The debates about the role of the government in education investments usually center on the economic benefits of the policy. I show that economic benefits to households depend on a few crucial factors: the costs of education, the labor market returns to education, and, importantly, the GE changes in earnings. While these are sufficient in capturing the direct economic benefits, education can have other welfare consequences as well, such as better health or more informed political participation (Sen 1999; Clark and Royer 2013). Exploring these relationships is left for future research.

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